Your Cancer Summary Report

Patient 001

Stage IIIC High-Grade Serous Ovarian Carcinoma

Precision Medicine Research Center

Report Date: March 09, 2026

⚠️ **DRAFT - FOR CLINICIAN REVIEW**

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Ovarian Cancer - High-Grade Serous Carcinoma (HGSOC)

Stage: Stage IIIC | **Grade:** Grade 3

You have been diagnosed with an aggressive form of ovarian cancer. Our research team analyzed your tumor using advanced spatial genomics and immune profiling to identify the treatments most likely to help you.

✓ Positive Factors

- CD8+ T cells present (9.1%) — can be reactivated with immunotherapy
- Strong neoantigen binder RMPEAAPPV confirmed (IC50=7.8nM)
- VEGFA strongly clustered — direct Bevacizumab target

⚠ Challenges to Consider

- 78% immune evasion rate
- High stromal content may create drug-resistant barrier

This AI-generated summary must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice.

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

TP53

Likely Pathogenic

Likely pathogenic (HGSOC hallmark)

A gene change found in nearly all HGSOC tumors — a key driver of cancer aggressiveness.

Treatment Option: Diagnostic marker; research-stage targeted therapies in development

BRCA1/HRD

VUS — formal testing required

HRD-positive profile suspected

Your tumor shows DNA repair defects. If confirmed, PARP inhibitors would be highly effective.

Treatment Option: Olaparib/Niraparib FDA-approved for HRD+ HGSOC

VEGFA

Likely Pathogenic

Spatially clustered (Moran's I=0.092, p<0.0001)

Blood vessel growth protein highly active in necrotic/hypoxic regions. Direct target for Bevacizumab.

Treatment Option: Bevacizumab FDA-approved for first-line ovarian cancer

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Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment
900 Visium spots, 31-gene panel. 6 regions: tumor_core=69, tumor_proliferative=124, tumor_interface=112, stroma_immune=212, stroma=180, necrotic_hypoxic=203. VEGFA highest in necrotic/hypoxic region (555 TPM). CD8A+CD68+FOXP3 all highest in stroma_immune.

Immune Status: ● Mixed immune activity - Variable immune response across the tumor

Your tumor has 6 distinct zones. The proliferating tumor zone has the highest cancer cell activity (EPCAM=1904). The immune zone shows both T cells and macrophages but most are suppressed. Blood vessel growth is highest in the oxygen-starved zone.

Key Patterns Observed

- VEGFA: Moran's I=0.092 — strongly clustered in necrotic/hypoxic region
- CD8A: 493 TPM in stroma_immune vs 65-89 elsewhere
- EPCAM: 1904 TPM in tumor_proliferative — highest proliferating tumor signal
- 78% immune cell evasion by quantum analysis

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Treatment Options

Based on your test results, your care team may consider these treatments:

Carboplatin + Paclitaxel

chemotherapy

NCCN Category 1

Standard first-line chemotherapy. Given by IV every 3 weeks for 6 cycles. Stops cancer cells dividing.

Why this may help you: Standard of care for Stage IIIC HGSOC.

Common side effects: Hair loss, Nausea, Fatigue, Low blood counts

Bevacizumab (Avastin)

targeted

FDA Approved

Cuts off blood supply to tumor by blocking VEGF. Given by IV alongside chemotherapy.

Why this may help you: VEGFA most spatially clustered gene (Moran's I=0.092). Highest expression in necrotic/hypoxic regions. Strong molecular match.

Common side effects: High blood pressure, Protein in urine

Olaparib / Niraparib

targeted

FDA Approved

Oral pills that exploit your tumor DNA repair defect. Highly effective in HRD+ patients.

Why this may help you: HRD-positive profile suspected. Formal Myriad myChoice testing urgently recommended.

Common side effects: Nausea, Fatigue, Low blood counts

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Pembrolizumab

immunotherapy

NCCN Category 2B

Removes immune brakes so your T cells can attack the tumor again.

Why this may help you: 78% immune evasion, CD8+ T cells present at 9.1%, RMPEAAPPV confirmed strong binder IC50=7.8nM.

Common side effects: Immune inflammation, Fatigue, Rash

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 3 months CA-125 blood test
Track tumor activity

Every 3-6 months CT scan (chest, abdomen, pelvis)
Assess tumor response

Once — urgent HRD/BRCA formal testing
Confirm PARP inhibitor eligibility

⚠️ When to Call Your Doctor

Contact your care team right away if you experience:

- Sudden severe abdominal pain
- Unexplained weight loss >10 lbs
- Fever above 101F during chemotherapy

Questions? Your oncology team or nurse navigator

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⚠️ **Important Notice**

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